

APPENDIX

A. Error analysis

As can be seen from Algorithm ?? (DP-CRS), the generation of the final synthetic dataset $\tilde{\mathcal{D}}$ is determined by the weights $w^{(T)}$ of each sample in the candidate set $\mathcal{C}$, and these weights gradually fit the true distribution of the private dataset $\mathcal{D}_{\text{priv}}$ through T rounds of iterations. Specifically, the weight update in each round depends on the noisy prefix sum $\tilde{Z}^{(t)}$ released by the privacy-preserving mechanism (TreeAggregator), which directly determines the direction and magnitude of the weight update. Therefore, in conducting the overall error analysis, we first measure the bias and error introduced by the noisy prefix sum $\tilde{Z}^{(t)}$ compared with the true utility signal (i.e., the true gradient $\mu = \mathbb{E}[g^{(t)}]$), and then substitute this error term into the weight iterative update process to further derive and calculate the final overall error of the algorithm.

Further, we observe that the noisy prefix sum $\tilde{Z}^{(t)}$ at each iteration of the algorithm is generated from the empirical utility signal $g^{(t)}$ of the corresponding round, with noise subsequently injected through the tree-aggregation privacy mechanism. Based on this generation pipeline and the empirical risk decomposition framework in statistical learning theory, we decompose the total error of $\tilde{Z}^{(t)}$ relative to the true utility signal μ into two independently analyzable components: the empirical estimation error, and the stochastic noise perturbation from the differential privacy mechanism. The specific formulation of each error term is presented as follows:

$$\tilde{Z}^{(t)} = \underbrace{\mu}_{\text{true sigal}} + \underbrace{e_{\text{est}}^{(t)}}_{\text{estimating}} + \underbrace{e_{\text{DP}}^{(t)}}_{\text{noise}}$$

Next, we first rigorously derive and upper-bound each of the three aforementioned error components separately, and then substitute them into the exponential refinement iteration process to derive and quantify the overall cumulative error introduced by the weight update procedure.

1) Error from estimating:

Lemma A.1. *Let $\delta > 0$. Then with probability at least $1 - \delta$, the sampling error*

$$e_{\text{est}}^{(t)} = \left\| \frac{1}{m} \sum_{j=1}^m g^{(t)}(x_j) - \mathbb{E}[g^{(t)}] \right\|_2 \leq \sqrt{\frac{2 \log(1/\delta)}{m}}$$

for any iteration t .

Lemma A.2 (McDiarmid's Inequality). *Let $X_1, X_2, \dots, X_n$ be independent random variables taking values in sets $\mathcal{X}_1, \mathcal{X}_2, \dots, \mathcal{X}_m$, respectively. Suppose a function $f : \prod_{i=1}^m \mathcal{X}_i \rightarrow \mathbb{R}$ satisfies the bounded difference property: there exist constants $c_1, c_2, \dots, c_n \geq 0$ such that for all $i \in [m]$,*

$$\sup_{x_1, \dots, x_m, x'_i} |f(x_1, \dots, x_i, \dots, x_m) - f(x_1, \dots, x'_i, \dots, x_m)| \leq c_i,$$

where the supremum is taken over all $x_j \in \mathcal{X}_j$ ($j \neq i$) and $x_i, x'_i \in \mathcal{X}_i$. Then for any $\lambda > 0$,

$$\Pr[f(X_1, \dots, X_m) - \mathbb{E}[f(X_1, \dots, X_m)] \geq \lambda] \leq \exp\left(-\frac{2\lambda^2}{\sum_{i=1}^n c_i^2}\right).$$

Lemma A.3 (Jensen Inequality). *For any convex function $\phi(\cdot)$ and integrable random variable X , we have $\phi(\mathbb{E}[X]) \leq \mathbb{E}[\phi(X)]$.*

In particular, for the ℓ_2 -norm (which is convex), it holds that $\|\mathbb{E}[X]\|_2 \leq \mathbb{E}[\|X\|_2]$; for a non-negative random variable X , we have $(\mathbb{E}[X])^2 \leq \mathbb{E}[X^2]$.

Proof. of Lemma A.1

Let $X_j := g^{(t)}(x_j) \in \mathbb{R}^N$ for each sample x_j in the private dataset at iteration t . By the algorithm design, $X_1, X_2, \dots, X_m$ are independent and identically distributed (i.i.d.) random vectors sampled from the true data population, and satisfy $\|X_j\|_2 \leq 1$ for all $j \in [m]$ according to the bound of gradient.

Denote the empirical mean as $\bar{X} = \frac{1}{m} \sum_{j=1}^m X_j$ and the true overall expectation as $\mu = \mathbb{E}[X_j]$. We aim to bound the ℓ_2 error between the empirical mean and true expectation, i.e., $\|\bar{X} - \mu\|_2$.

First, we define the function:

$$f(X_1, X_2, \dots, X_m) = \left\| \frac{1}{m} \sum_{j=1}^m X_j - \mu \right\|_2$$

For any $j \in [m]$, replacing X_j with X'_j (which also satisfies $\|X'_j\|_2 \leq 1$), by the reverse triangle inequality of the ℓ_2 norm, we have:

$$\left| f(\dots, X_j, \dots) - f(\dots, X'_j, \dots) \right| \leq \frac{1}{m} \|X_j - X'_j\|_2 \leq \frac{2}{m}.$$

Thus, f satisfies the bounded difference condition with constant $c_j = \frac{2}{m}$ for all $j \in [m]$.

By the one-sided McDiarmid's inequality (Lemma A.2), for any $\lambda > 0$, we have:

$$\Pr(f - \mathbb{E}[f] \geq \lambda) \leq \exp\left(-\frac{2\lambda^2}{\sum_{j=1}^m c_j^2}\right) = \exp\left(-\frac{m\lambda^2}{2}\right).$$

Setting the right-hand side equal to δ solves to:

$$\lambda = \sqrt{\frac{2 \log(1/\delta)}{m}}.$$

Therefore, with probability at least $1 - \delta$, it holds that:

$$f \leq \mathbb{E}[f] + \sqrt{\frac{2 \log(1/\delta)}{m}}.$$

Next, we bound $\mathbb{E}[f]$ via Jensen's inequality (Lemma A.3). We first expand $\mathbb{E}[f^2]$ using the independence of X_j :

$$\begin{aligned} \mathbb{E}[f^2] &= \mathbb{E}[\|\bar{X} - \mu\|_2^2] = \mathbb{E}\left[\left\| \frac{1}{m} \sum_{j=1}^m (X_j - \mu) \right\|_2^2\right] \\ &= \frac{1}{m^2} \sum_{j=1}^m \mathbb{E}[\|X_j - \mu\|_2^2], \end{aligned}$$

where cross terms vanish due to zero covariance between independent random variables. Since $\|X_j - \mu\|_2 \leq 2$, we have $\mathbb{E}[\|X_j - \mu\|_2^2] \leq 4$, which gives:

$$\mathbb{E}[f^2] \leq \frac{4}{m}.$$

By Jensen's inequality $\mathbb{E}[f]^2 \leq \mathbb{E}[f^2]$, we obtain:

$$\mathbb{E}[f] \leq \frac{2}{\sqrt{m}}.$$

Substituting (2) into (1), with probability at least $1 - \delta$, we have:

$$\|\bar{X} - \mu\|_2 \leq \frac{2}{\sqrt{m}} + \sqrt{\frac{2 \log(1/\delta)}{m}}.$$

For the standard setting in differential privacy where $\delta \ll 1$, we can rigorously absorb the constant term to get the concise bound:

$$\left\| \frac{1}{m} \sum_{j=1}^m g^{(t)}(x_j) - \mathbb{E}[g^{(t)}] \right\|_2 \leq \sqrt{\frac{2 \log(1/\delta)}{m}},$$

which completes the proof. $\square$

2) *Error from DP Noise:* We use a tree-based cumulative publishing mechanism in the algorithm to reduce noise accumulation by publishing interval statistical results. In error analysis, we use the worst-case error guarantee, namely the standard combinatorial theorem, for the analysis.

Lemma A.4. *Let $\delta > 0$. For any iteration t , let the released noisy signal be $\tilde{Z}^{(t)} = \sum_{s \in t} \tilde{g}^{(s)} + \log t \cdot \xi^{(t)}$, where $\xi^{(t)} \sim \mathcal{N}(0, \sigma_{\text{tree}}^2)$ with $\sigma^2 = \frac{2T}{m^2\rho}$. Then, with probability at least $1 - \delta$, the noise-induced estimation error satisfies*

$$e_{\text{noise}}^{(t)} = \left\| \tilde{Z}^{(t)} - \sum_{s \in t} \tilde{g}^{(s)} \right\|_2 \leq \frac{\log t}{m} \sqrt{4TN \log(1/\delta)}.$$

Proof. By definition, the noise error is exactly the ℓ_2 -norm of the Gaussian noise vector:

$$e_{\text{noise}}^{(t)} = \left\| \tilde{Z}^{(t)} - \sum_{s \in t} \tilde{g}^{(s)} \right\|_2 \leq \log t \cdot \|\xi^{(t)}\|_2.$$

Let $\xi^{(t)} \in \mathbb{R}^N$ be i.i.d. Gaussian noise with each entry $\xi_i \sim \mathcal{N}(0, \sigma^2)$. Define standardized noise $\eta_i = \xi_i / \sigma \sim \mathcal{N}(0, 1)$. Then

$$\|\xi^{(t)}\|_2 = \sigma \cdot \sqrt{\sum_{i=1}^N \eta_i^2}.$$

Using the standard concentration bound for the squared norm of a standard Gaussian vector:

$$\mathbb{P}\left(\|\eta\|_2^2 \geq 2N \log \frac{1}{\delta}\right) \leq \delta.$$

Rearranging gives

$$\mathbb{P}\left(\|\xi^{(t)}\|_2 \leq \sigma \sqrt{2N \log \frac{1}{\delta}}\right) \geq 1 - \delta,$$

which completes the proof. $\square$

3) Global error:

Theorem A.1 (Convergence Error Bound of DP-CRS). *Let $\mu = \mathbb{E}[g^{(t)}]$ be the true expected utility signal, and let $w^* \in \Delta_N$ denote the optimal probability distribution over the candidate set that maximizes the expected utility $w^* = \arg \max_{w \in \Delta_N} \langle w, \mu \rangle$, where $\Delta_N = \{w \in \mathbb{R}_{\geq 0}^N : \sum_{i=1}^N w_i = 1\}$ is the probability simplex. Let $w^{(t)}$ the weight vector at the t -th iteration, $g^{(t)}$ the true utility vector, N the size of the candidate set, m the number of samples in the private dataset, and T the total number of iteration rounds. Then, with probability at least $1 - \delta$, the average expected regret of the learned weight sequence satisfies*

$$\frac{1}{T} \sum_{t=1}^T \langle w^* - w^{(t)}, \mathbb{E}[g] \rangle \leq O\left(\sqrt{\frac{\log N}{T}} + \frac{1}{\sqrt{m}} + \frac{\sqrt{NT}}{m\sqrt{\rho}}\right).$$

Lemma A.5 (Hölder's Inequality). *Let $p, q \in [1, \infty]$ be conjugate exponents satisfying $\frac{1}{p} + \frac{1}{q} = 1$. For any vectors $a, b \in \mathbb{R}^N$, we have*

$$|\langle a, b \rangle| \leq \|a\|_p \|b\|_q.$$

In particular, for the ℓ_1 - ℓ_∞ pair ($p = 1, q = \infty$), it holds that

$$|\langle a, b \rangle| \leq \|a\|_1 \|b\|_\infty.$$

Proof. Based on the above derivations, we have completed the upper bound analysis for the three core error components of the noisy prefix sum $\tilde{Z}^{(t)} = \mathbb{E}[g^{(t)}] + e_{\text{est}}^{(t)} + e_{\text{DP}}^{(t)}$. On this basis, we substitute the above error terms into the cumulative weight refinement iteration process of the DP-CRS algorithm, to quantitatively analyze the end-to-end overall convergence error between the learned weight sequence after multiple rounds of exponential gradient descent iterations and the optimal weight distribution over the candidate set.

At iteration t , the algorithm updates the weights as

$$\log w_i^{(t)} \leftarrow \log w_i^{(t-1)} + \eta \tilde{Z}_i^{(t)}, \quad \forall i \in [N],$$

where $\tilde{Z}^{(t)} = (\tilde{Z}_1^{(t)}, \dots, \tilde{Z}_N^{(t)})$ is the noisy prefix sum vector released by the TreeAggregator mechanism. This shows that the proposed update rule can be interpreted as an Exponentiated Gradient (EG) procedure over the N -dimensional probability simplex $\Delta_N = \{w \in \mathbb{R}_{\geq 0}^N : \sum_{i=1}^N w_i = 1\}$.

To analyze the convergence of the update rule, we adopt the mirror descent framework with the negative entropy as the mirror map, i.e., $\psi(w) = \sum_{i=1}^N w_i \log w_i$, whose induced Bregman divergence is exactly the Kullback-Leibler (KL) divergence $D_{KL}(w\|v) = \sum_{i=1}^N w_i \log \frac{w_i}{v_i}$. This framework naturally aligns with the EG update, which operates on the probability simplex. At each iteration t , the algorithm uses the noisy prefix sum $\tilde{Z}^{(t)}$ as a surrogate gradient of the true expected utility vector $\mu = \mathbb{E}[g^{(t)}]$ to update the weight distribution $w^{(t)}$.

Under this formulation, we measure the algorithm's performance by the expected regret, which compares the cumulative expected utility achieved by the learned weight sequence

$\{w^{(t)}\}_{t=1}^T$ against that of the best fixed optimal distribution $w^* \in \Delta_N$ in hindsight. Formally, the regret is defined as

$$\text{Regret}(T) = \sum_{t=1}^T \langle w^* - w^{(t)}, \mu \rangle,$$

where $w^* = \arg \max_{w \in \Delta_N} \langle w, \mu \rangle$ is the optimal distribution over the candidate set that maximizes the expected utility. The term $\langle w^* - w^{(t)}, \mu \rangle$ quantifies the suboptimality of the learned weight $w^{(t)}$ relative to w^* under the true expected utility $\mu = \mathbb{E}[g^{(t)}]$ at iteration t . Minimizing this regret ensures that the learned distribution $w^{(t)}$ progressively converges to the optimal distribution w^* , even with noisy gradient estimates.

We now derive the regret bound via the standard mirror descent regret guarantee for KL divergence [1]. For the EG update with learning rate η , the regret with respect to the noisy surrogate gradient $\tilde{Z}^{(t)}$ satisfies:

$$\sum_{t=1}^T \langle w^* - w^{(t)}, \tilde{Z}^{(t)} \rangle \leq \frac{D_{KL}(w^* \| w^{(0)})}{\eta} + \eta \sum_{t=1}^T \|\tilde{Z}^{(t)}\|_\infty^2.$$

For our algorithm, the initial weight $w^{(0)}$ is the uniform distribution over the candidate set, so $D_{KL}(w^* \| w^{(0)}) \leq \log N$ for any $w^* \in \Delta_N$. Substituting this into the bound gives:

$$\sum_{t=1}^T \langle w^* - w^{(t)}, \tilde{Z}^{(t)} \rangle \leq \frac{\log N}{\eta} + \eta \sum_{t=1}^T \|\tilde{Z}^{(t)}\|_\infty^2. \quad (1)$$

Next, we substitute the gradient decomposition $\tilde{Z}^{(t)} = \mu + e_{\text{est}}^{(t)} + e_{\text{DP}}^{(t)}$ into the left-hand side of 1:

$$\begin{aligned} \sum_{t=1}^T \langle w^* - w^{(t)}, \mu \rangle + \sum_{t=1}^T \langle w^* - w^{(t)}, e_{\text{est}}^{(t)} + e_{\text{DP}}^{(t)} \rangle \\ \leq \frac{\log N}{\eta} + \eta \sum_{t=1}^T \|\tilde{Z}^{(t)}\|_\infty^2. \end{aligned}$$

Rearranging terms, we obtain the bound for the target expected regret:

$$\begin{aligned} \sum_{t=1}^T \langle w^* - w^{(t)}, \mu \rangle &\leq \frac{\log N}{\eta} + \eta \sum_{t=1}^T \|\tilde{Z}^{(t)}\|_\infty^2 \\ &\quad - \sum_{t=1}^T \langle w^* - w^{(t)}, e_{\text{est}}^{(t)} + e_{\text{DP}}^{(t)} \rangle. \end{aligned} \quad (2)$$

We now bound the error terms on the right-hand side of (2) one by one. For any $w^*, w^{(t)} \in \Delta_N$, we have $\|w^* - w^{(t)}\|_1 \leq 2$ by the definition of the probability simplex. By Lemma A.5 (ℓ_1 - ℓ_∞), for any error vector $e^{(t)}$, we have:

$$\left| \langle w^* - w^{(t)}, e^{(t)} \rangle \right| \leq \|w^* - w^{(t)}\|_1 \cdot \|e^{(t)}\|_\infty \leq 2\|e^{(t)}\|_\infty.$$

Alternatively, by the Cauchy-Schwarz inequality, we have:

$$\left| \langle w^* - w^{(t)}, e^{(t)} \rangle \right| \leq \|w^* - w^{(t)}\|_2 \cdot \|e^{(t)}\|_2 \leq \sqrt{2}\|e^{(t)}\|_2,$$

which aligns with the ℓ_2 bound of our previously derived error terms. We use the ℓ_2 form below to match our previous lemmas.

Bound the empirical estimation error. By Lemma A.1, for any iteration t , with probability at least $1 - \delta_1$, we have:

$$\|e_{\text{est}}^{(t)}\|_2 \leq \sqrt{\frac{2 \log(T/\delta_1)}{m}}.$$

We replace δ with δ_1/T to ensure the bound holds for all T iterations via the union bound. Summing over $t = 1$ to T , we get:

$$\left| \sum_{t=1}^T \langle w^* - w^{(t)}, e_{\text{est}}^{(t)} \rangle \right| \leq \sqrt{2} \sum_{t=1}^T \|e_{\text{est}}^{(t)}\|_2 \leq T \cdot \sqrt{\frac{4 \log(T/\delta_1)}{m}}. \quad (3)$$

Bound the DP noise perturbation. By Lemma A.4, for any iteration t , with probability at least $1 - \delta_2$, we have:

$$\|e_{\text{DP}}^{(t)}\|_2 \leq \frac{\log T}{m} \sqrt{\frac{4TN \log(1/\delta)}{\rho}}.$$

We replace δ with δ_2/T to ensure the bound holds for all T iterations via the union bound. Summing over $t = 1$ to T , we get:

$$\left| \sum_{t=1}^T \langle w^* - w^{(t)}, e_{\text{DP}}^{(t)} \rangle \right| \leq \sqrt{2} \sum_{t=1}^T \|e_{\text{DP}}^{(t)}\|_2 \leq \frac{T \log T}{m} \sqrt{\frac{8TN \log(1/\delta_2)}{\rho}}. \quad (4)$$

Bound the gradient norm term. Under the gradient's bound $\|\tilde{g}^{(t)}\|_2 \leq 1$, and with the DP noise bounded by a constant G_σ , we have $\|\tilde{Z}^{(t)}\|_\infty \leq G = 1 + G_\sigma$ for a universal constant G (since softmax outputs are in $[0, 1]$ and noise is bounded with high probability). Thus:

$$\eta \sum_{t=1}^T \|\tilde{Z}^{(t)}\|_\infty^2 \leq \eta T G^2. \quad (5)$$

We now provide a coarse upper bound for G_σ to further characterize the magnitude of G .

We give the high-probability bound for the per-iteration noise norm, which aligns with the standard concentration inequality for Gaussian vectors and the union bound for T iterations:

Lemma A.6 (Per-iteration DP Noise Bound). *Let $\delta > 0$. For any iteration $t \in [T]$, with probability at least $1 - \delta/T$, the ℓ_2 norm of the tree-aggregation noise satisfies*

$$\|\xi^{(t)}\|_2 \leq \sqrt{\frac{2C^2 \log T}{m^2 \rho}} \cdot \left(\sqrt{2 \log(T/\delta)} + \sqrt{N} \right).$$

Proof. Each coordinate of $\xi^{(t)}$ follows an independent Gaussian distribution $\mathcal{N}(0, \sigma_{\text{tree}}^2)$, so $\xi^{(t)}$ is an N -dimensional Gaussian vector with covariance matrix $\sigma_{\text{tree}}^2 I_N$. By the standard concentration inequality for the ℓ_2 norm of Gaussian vectors, for any $\gamma > 0$, we have

$$\Pr \left(\|\xi^{(t)}\|_2 \geq \sigma_{\text{tree}} \cdot (\sqrt{N} + \gamma) \right) \leq \exp \left(-\frac{\gamma^2}{2} \right).$$

For typical settings where $\log N < \log(T/\delta)$, we can absorb the dimension term into the logarithmic term, and set

$\gamma = \sqrt{2 \log(T/\delta)}$ to make the right-hand side equal to δ/T . Substituting $\sigma_{\text{tree}} = \sqrt{\frac{2C^2 \log T}{m^2 \rho}}$ completes the proof. $\square$

Finally, we get:

$$\begin{aligned} \eta \sum_{t=1}^T \|\tilde{Z}^{(t)}\|_\infty^2 &\leq \eta T G^2 \\ &= \eta T \left(\sqrt{\frac{2C^2 \log T}{m^2 \rho}} \left(\sqrt{2 \log(T/\delta)} + \sqrt{N} \right) + 1 \right)^2. \end{aligned} \quad (6)$$

Combine all bounds and optimize the learning rate.

Substitute (3), (4), (6) into (2). For the bounded utility setting where $\|g_{\text{emp}}^{(t)}\|_2 \leq 1$, and setting $\delta_1 = \delta_2 = \delta/2$ via the union bound to ensure the overall probability guarantee is at least $1 - \delta$, we obtain:

$$\begin{aligned} \sum_{t=1}^T \langle w^* - w^{(t)}, \mu \rangle &\leq \frac{\log N}{\eta} + \eta T G^2 \\ &\quad + T \sqrt{\frac{4 \log(2T/\delta)}{m}} \\ &\quad + \frac{T \log T}{m} \sqrt{\frac{8 T N \log(1/\delta)}{\rho}}. \end{aligned}$$

To minimize the bound, we optimize the learning rate η by balancing the first two terms. Taking the derivative with respect to η and setting it to zero gives the optimal learning rate:

$$\eta^* = \sqrt{\frac{\log N}{T G^2}}.$$

Substituting η^* into the bound, the first two terms become $2G\sqrt{T \log N}$.

Finally, we take the average over T iterations to get the average expected regret bound. Absorbing all constants into the big-O notation, we obtain the final convergence result:

$$\frac{1}{T} \sum_{t=1}^T \langle w^* - w^{(t)}, \mathbb{E}[g^{(t)}] \rangle \leq O \left(\sqrt{\frac{\log N}{T}} + \frac{1}{\sqrt{m}} + \frac{\sqrt{NT}}{m\sqrt{\rho}} \right),$$

which holds with probability at least $1 - \delta$. $\square$

B. Feasibility Analysis of Tree Structure

In conventional iterative differential privacy mechanisms, the utility statistic at each round is released independently. Let

$$\bar{g}^{(t)} = \frac{1}{m} \sum_{j=1}^m g^{(t)}(x_j)$$

denote the aggregated utility vector computed at iteration t . A standard approach perturbs each $\bar{g}^{(t)}$ with Gaussian noise and releases the noisy statistic for subsequent optimization. Since the utility at round t depends on the previously released outputs through the weight update process, the sequence $\{\bar{g}^{(t)}\}_{t=1}^T$ forms a collection of adaptive queries over the

same private dataset. Consequently, the privacy guarantee is typically established through standard composition, where the total privacy budget accumulates over all T rounds. Under this formulation, each iteration is treated as an independent DP release, and the privacy analysis naturally scales with the number of optimization rounds.

Our method differs fundamentally in that it does not treat the utility statistics as isolated releases. Instead, we define a cumulative utility signal

$$Z^{(t)} = \sum_{s=1}^t \bar{g}^{(s)},$$

which aggregates the utility information observed throughout the optimization process. The weight refinement procedure is therefore driven by a cumulative view of the utility sequence rather than a collection of independent per-round statistics. This cumulative structure introduces a natural prefix-sum representation over time, transforming the release problem from repeatedly publishing individual utilities into maintaining and releasing a sequence of cumulative utility signals $\{Z^{(t)}\}_{t=1}^T$.

The cumulative formulation enables the use of a binary tree aggregation structure. Specifically, a complete binary tree is constructed over the T optimization rounds, where each node corresponds to a contiguous time interval. For a node v covering interval $I(v)$, we define its interval statistic as

$$S_v = \sum_{s \in I(v)} \bar{g}^{(s)}.$$

A noisy node release is obtained by

$$\tilde{S}_v = S_v + \xi_v, \quad \xi_v \sim \mathcal{N}(0, \sigma^2 I).$$

At any iteration t , the prefix interval $[1, t]$ can be uniquely decomposed into a collection of disjoint canonical nodes $C(t)$. Therefore, the cumulative utility signal can be reconstructed as

$$\tilde{Z}^{(t)} = \sum_{v \in C(t)} \tilde{S}_v = Z^{(t)} + \sum_{v \in C(t)} \xi_v.$$

Since the canonical decomposition contains only $O(\log T)$ nodes, each cumulative release depends on a logarithmic number of noisy interval statistics rather than a linear number of independently perturbed rounds. Consequently, the tree structure provides an efficient mechanism for maintaining cumulative utility signals and substantially mitigates the accumulation of reconstruction error over long optimization horizons. In contrast, conventional iterative releases maintain no such cumulative representation and therefore rely solely on standard composition across all optimization rounds.

It is important to note that the tree structure is introduced to maintain and reconstruct cumulative utility signals, rather than assuming the continual-observation setting in which each data record affects only a single time step. In our framework, the utility sequence is generated through an adaptive optimization process on a fixed private dataset. Therefore, the role of the tree mechanism is to organize cumulative releases and reduce error accumulation in the reconstructed utility signal, while

the privacy analysis of the adaptive query sequence is still established through the composition of differentially private releases.

C. Missing Experimental Details

1) *Datasets*: The ADULT dataset is obtained from the 1994 US Census database, consisting of 14 attributes (6 numerical attributes and 8 categorical attributes) and approximately 48,842 individual tuples. The core task of it is to classify the "income" of adults.

The HIGGS dataset is a high-energy physics dataset generated via Monte Carlo simulations of proton-proton collisions at the Large Hadron Collider. It comprises 28 attributes (21 numerical attributes and 7 categorical attributes) and around 81,000 samples, and serves as a common benchmark for high-dimensional data evaluation.

The HTRU dataset collects pulsar candidate observations from radio astronomy surveys, containing 9 attributes (8 numerical attributes and 1 categorical attributes) and approximately 17,898 individual signal samples. It is a typical benchmark for imbalanced classification and synthetic data validation.

The kdd-cup10 dataset originates from the "Educational Data Mining Challenge" held at the 2010 ACM SIGKDD International Conference. This dataset contains approximately 8 million interaction records between students and the tutoring system, involving 12 core attributes, including 6 numerical attributes and 6 categorical attributes. The core task of this subset is to predict the correctness of students' answers to questions.

The Shoppers dataset (full name: Online Shopper Behavior Dataset) is a benchmark dataset for shopping behavior focusing on the online retail scenario, containing 12,330 independent user session records. This dataset includes 10 numerical attributes and 8 categorical attributes, with the core target variable being "Revenue" (marking whether a session results in a purchase).

REFERENCES

- [1] S. Shalev-Shwartz, "Online learning and online convex optimization," *Foundations and Trends® in Machine Learning*, vol. 4, no. 2, pp. 107–194, 2025.